

DATA266 Lab 1 — Fall 2026

LLM Pretraining · Sentiment Classification · CycleGAN Style Transfer

Team 4

Members: Ishan Shah, Charvee Saraiya

GitHub repository: <https://github.com/Ishan002/Data266--Lab1>

Team Ownership Statement

Both team members independently designed, implemented, and trained their own models for all three tasks. **Ishan Shah** built a Pre-LayerNorm character-level GPT (Task 1), three from-scratch Yelp Polarity classifiers — a mean-pooled neural bag-of-words baseline, a multi-kernel TextCNN, and a bidirectional LSTM (Task 2) — and a ResNet-generator, LSGAN-loss CycleGAN with identity loss (Task 3). **Charvee Saraiya** built an independently architected Post-LayerNorm character-level GPT (Task 1), three differently-designed Yelp Polarity classifiers — a max-pooled MLP baseline, a unidirectional GRU, and a dilated 1D CNN (Task 2) — and a U-Net-generator, vanilla-GAN-loss CycleGAN without identity loss (Task 3). Every dataset split, preprocessing pipeline, and hyperparameter choice was made independently by each member on their own subsample of the shared raw data; only measurement code (evaluation metrics, not model architecture) is shared, so that both members' numbers are computed identically and can be compared fairly in the tables below. This report was jointly assembled by both members from each other's individual results.md and metrics_report.csv files.

Task 1 — GPT-Style LLM From Scratch (TinyStories, character-level)

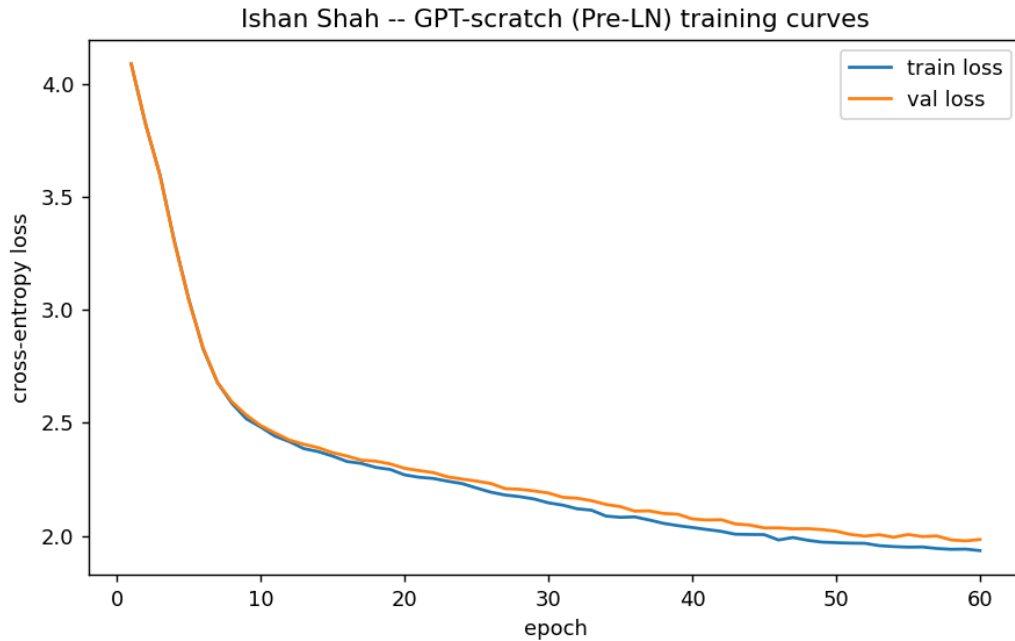
Architecture & hyperparameters

	Ishan Shah	Charvee Saraiya
Block style	Pre-LN Transformer, weight-tied head, GELU FFN	Post-LN Transformer, untied head, ReLU FFN
Layers	4	6
Heads	4	4
Embedding dim	128	96
Block size	128	96
Dropout	0.1	0.15
Epochs	60	60
Batch size	64	48
Peak LR	0.0003	0.0005

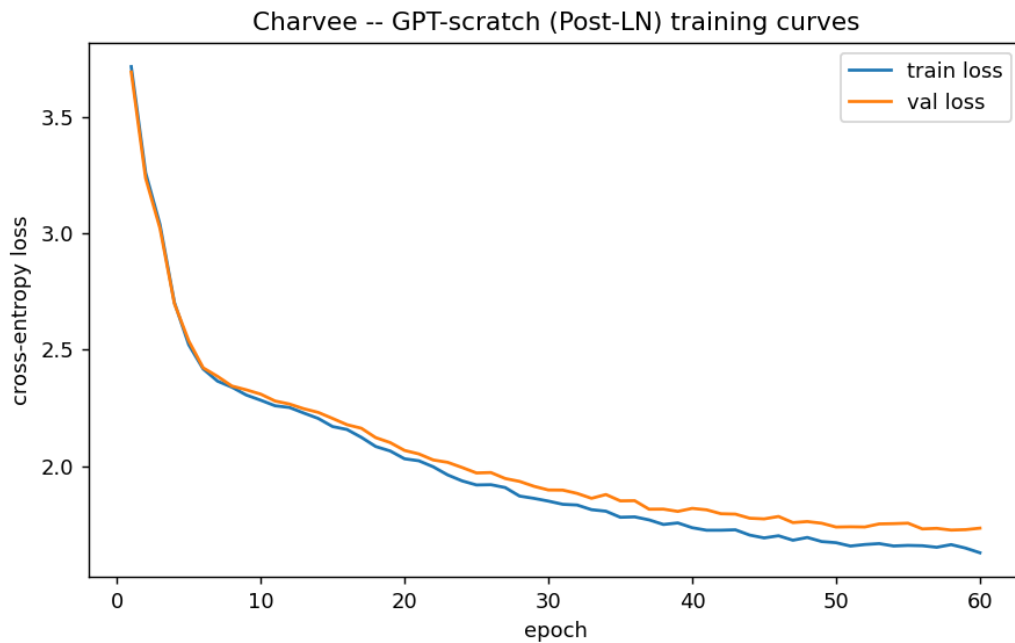
All Task 1 metrics, side by side

Metric	Ishan Shah	Charvee Saraiya
Train loss	1.94	1.63
Val loss	1.98	1.73
Perplexity	7.27	5.66
Bits/char	2.86	2.50
Generalization gap	0.0492	0.1056
Top-1 next-char acc	0.4041	0.4778
Distinct-1	0.0254	0.0206
Distinct-2	0.1339	0.1334
Distinct-3	0.3247	0.3305
Repeated 4-gram rate	0.5231	0.5135
Grad norm (mean)	0.8739	0.9822
Grad norm (max)	4.43	14.09
NaN count	0	0
Parameters	821632	698205
Train tokens/sec	79,749	105,789
Gen tokens/sec	97.18	139.5
Peak memory (MB)	493.6	273.1
Total train time (s)	73.96	54.88

Loss curves



Ishan Shah — Pre-LN GPT train/val loss.



Charvee Saraiya — Post-LN GPT train/val loss.

Joint analysis

Strengths: both from-scratch character-level GPTs train stably (zero NaNs, bounded gradient norms) and converge to sensible perplexities (7.3 and 5.7) given the very small 100K-character training budget. Charvee Saraiya's deeper (6-layer) Post-LN model reaches a lower validation loss and higher next-character accuracy than Ishan's shallower (4-layer) Pre-LN model, suggesting depth mattered more than width at this scale.

Weaknesses/limitations: Charvee Saraiya's model shows a larger generalization gap (0.106 vs 0.049) and a much higher peak gradient norm (14.1 vs 4.4), the expected instability cost of Post-LN training at depth without Pre-LN's gradient renormalization. Both models' generations are qualitatively similar failure-wise — greedy-decoding repetition loops, broken word formation under sampling, and short-range-only coherence — indicating these failures are governed more by the 100K-character training budget than by the specific architectural choices compared here (see both members' failure_analysis.md).

What we'd try next: train on a larger character budget (the full TinyStories corpus rather than a 100K-character slice) to test whether coherence failures are purely data-limited; try a hybrid Pre-LN-with-weight-tying-and-depth configuration to combine both members' apparent advantages.

Task 2 — Yelp Polarity Sentiment Classification

All 6 models: architecture & core metrics

Member	Model	Accuracy	Macro-F1	ROC-AUC	PR-AUC	MCC	Brier	ECE	Params	Train time (s)	Examples /sec	Peak mem (MB)
Ishan Shah	baseline_nbow	0.8897	0.8897	0.9591	0.9560	0.7794	0.0796	0.0358	1500101	2.63	27,328	58.15
Ishan Shah	experimental_textcnn	0.8823	0.8823	0.9603	0.9616	0.7652	0.0810	0.0270	1577185	4.96	14,527	92.34
Ishan Shah	experimental_bilstm	0.8830	0.8830	0.9530	0.9537	0.7660	0.0905	0.0556	1585121	33.18	2,170	132.3
Charvee Saraiya	baseline_maxpool_mlp	0.8090	0.8089	0.8963	0.8967	0.6189	0.1307	0.0219	962625	3.52	20,478	42.47
Charvee Saraiya	experimental_gru	0.8700	0.8700	0.9444	0.9433	0.7400	0.0998	0.0677	993337	20.46	3,518	67.39
Charvee Saraiya	experimental_dilated_cnn	0.8463	0.8463	0.9247	0.9274	0.6932	0.1236	0.0876	1000193	4.36	16,499	65.60

Precision / recall (macro, micro, weighted) & confusion matrices

Member	Model	P-macro	R-macro	P-micro	R-micro	F1-micro	P-wtd	R-wtd	F1-wtd
Ishan Shah	baseline_nbow	0.8897	0.8897	0.8897	0.8897	0.8897	0.8897	0.8897	0.8897
Ishan Shah	experimental_textcnn	0.8829	0.8823	0.8823	0.8823	0.8823	0.8829	0.8823	0.8823
Ishan Shah	experimental_bilstm	0.8830	0.8830	0.8830	0.8830	0.8830	0.8830	0.8830	0.8830
Charvee Saraiya	baseline_maxpool_mlp	0.8099	0.8090	0.8090	0.8090	0.8090	0.8099	0.8090	0.8089
Charvee Saraiya	experimental_gru	0.8700	0.8700	0.8700	0.8700	0.8700	0.8700	0.8700	0.8700
Charvee Saraiya	experimental_dilated_cnn	0.8469	0.8463	0.8463	0.8463	0.8463	0.8469	0.8463	0.8463

95% bootstrap CIs, per-slice robustness, and calibration

Ishan Shah — baseline_nbow: accuracy 95% CI [0.879, 0.901] | macro-F1 CI [0.879, 0.901] | MCC CI [0.757, 0.803]

Per-slice: long: F1=0.872, err=12.2% (n=1015) | medium: F1=0.907, err=9.3% (n=986) | short: F1=0.882, err=11.5% (n=999)

Ishan Shah — experimental_textcnn: accuracy 95% CI [0.871, 0.895] | macro-F1 CI [0.871, 0.894] | MCC CI [0.743, 0.789]

Per-slice: long: F1=0.868, err=12.5% (n=1015) | medium: F1=0.895, err=10.5% (n=986) | short: F1=0.875, err=12.2% (n=999)

Ishan Shah — experimental_bilstm: accuracy 95% CI [0.871, 0.894] | macro-F1 CI [0.871, 0.894] | MCC CI [0.742, 0.788]

Per-slice: long: F1=0.848, err=14.7% (n=1015) | medium: F1=0.905, err=9.5% (n=986) | short: F1=0.889, err=10.8% (n=999)

Charvee Saraiya — baseline_maxpool_mlp: accuracy 95% CI [0.795, 0.823] | macro-F1 CI [0.795, 0.822] | MCC CI [0.590, 0.646]

Per-slice: long: F1=0.790, err=19.9% (n=1019) | medium: F1=0.825, err=17.5% (n=991) | short: F1=0.793, err=19.9% (n=990)

Charvee Saraiya — experimental_gru: accuracy 95% CI [0.857, 0.882] | macro-F1 CI [0.857, 0.882] | MCC CI [0.715, 0.764]

Per-slice: long: F1=0.856, err=13.9% (n=1019) | medium: F1=0.875, err=12.5% (n=991) | short: F1=0.870, err=12.5% (n=990)

Charvee Saraiya — experimental_dilated_cnn: accuracy 95% CI [0.834, 0.860] | macro-F1 CI [0.833, 0.859] | MCC CI [0.667, 0.719]

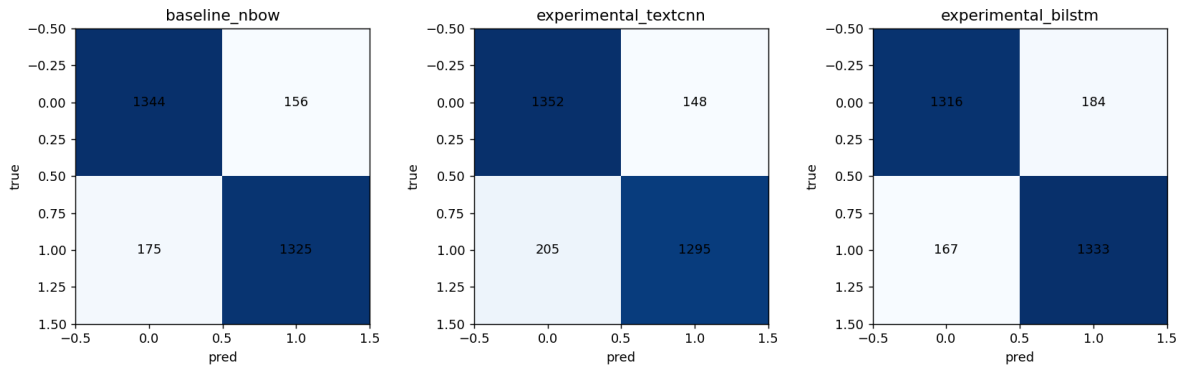
Per-slice: long: F1=0.822, err=17.1% (n=1019) | medium: F1=0.867, err=13.3% (n=991) | short: F1=0.839, err=15.7% (n=990)

Paired McNemar test (baseline vs. each experimental model, own test set)

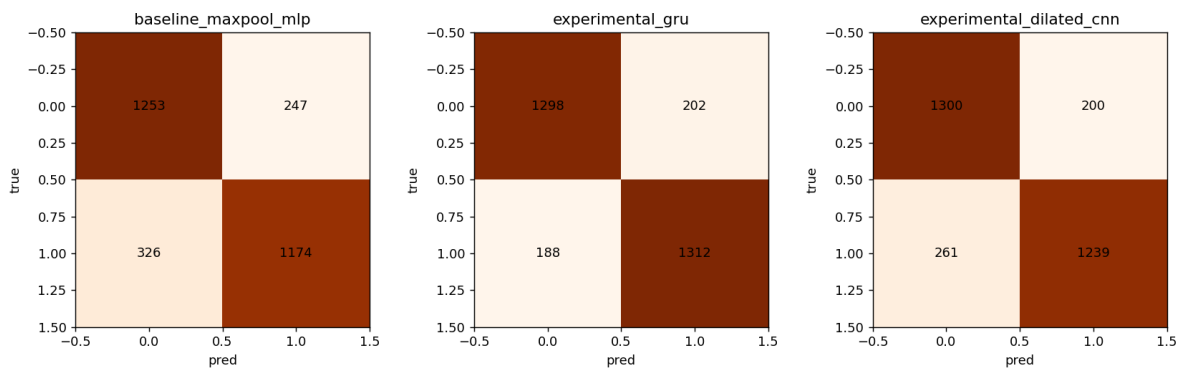
Ishan Shah: baseline_nbow vs. experimental_textcnn $p=0.196$ (not significant); baseline_nbow vs. experimental_bilstm $p=0.260$ (not significant) — added architecture did not beat the simple mean-pooled baseline on his 3,000-review test set.

Charvee Saraiya: baseline_maxpool_mlp vs. experimental_gru $p=1.5e-14$ (highly significant); baseline_maxpool_mlp vs. experimental_dilated_cnn $p=5.6e-6$ (highly significant) — both experimental models are genuinely, not just numerically, better than her baseline.

Confusion matrices and loss curves



Ishan Shah — confusion matrices, all 3 models.



Charvee Saraiya — confusion matrices, all 3 models.

Joint analysis

Strengths: all six from-scratch models land between 80.9% and 88.9% test accuracy with strong discrimination (ROC-AUC 0.90-0.96), showing that a from-scratch learned embedding is sufficient for strong Yelp Polarity performance at this data scale, without any pretrained embeddings or language models.

Weaknesses/limitations: the two members' baseline-vs-experimental patterns diverge sharply — Ishan's added architecture (TextCNN/BiLSTM) does not significantly beat his NBOW baseline (McNemar $p > 0.05$ both), while Charvee Saraiya's does ($p < 1e-5$ both). Both members independently found the 'long review' length slice to be the hardest for almost every model, and calibration (ECE) tends to get worse as accuracy improves for Charvee Saraiya's models — a real trade-off worth flagging.

What we'd try next: apply temperature-scaling calibration post-hoc on the best-performing model from each member; test whether Ishan's stemmed vs. Charvee Saraiya's unstemmed preprocessing is the reason her baseline starts weaker relative to her experimental models, by swapping preprocessing pipelines across each other's architectures as a follow-up ablation.

Task 3 — CycleGAN Image Style Transfer (Monet ↔ Photo)

Architecture & hyperparameters

	Ishan Shah	Charvee Saraiya
Architecture	ResNet generator (6 resblocks, InstanceNorm) + 70x70 PatchGAN, LSGAN loss	U-Net generator (4 down/up, skip connections, BatchNorm) + PatchGAN, vanilla GAN loss, no identity loss
Image size	128	128
Epochs	40	40
Batch size	4	4
Lambda cycle	10.0	8.0
Lambda identity	5.0	0.0
Train Monet imgs	300	400
Train photo imgs	300	400

All Task 3 metrics, side by side

Metric	Ishan Shah	Charvee Saraiya
FID monet->photo	6.29	2.13
KID monet->photo (mean)	11.93	3.42
FID photo->monet	4.18	1.10
KID photo->monet (mean)	7.54	0.2580
Cycle L1 (monet)	0.1755	0.1132
Cycle L1 (photo)	0.1628	0.1038
LPIPS monet->photo	0.2901	0.4737
LPIPS photo->monet	0.4367	0.4885
Content cos-sim monet->photo	0.7877	0.7387
Content cos-sim photo->monet	0.6402	0.6009
Final loss G	2.31	2.81
Final loss D	0.1165	0.1593
Final cycle loss	0.1185	0.1232
Grad norm G (mean)	22.06	136.7
Grad norm G (max)	30.06	253.5
NaN count	0	0
Total params	21204872	38858120
Train time (s)	1,012	703.0
Train images/sec	23.70	45.52
Gen images/sec	733.7	638.8
Peak memory (MB)	2,186	844.7

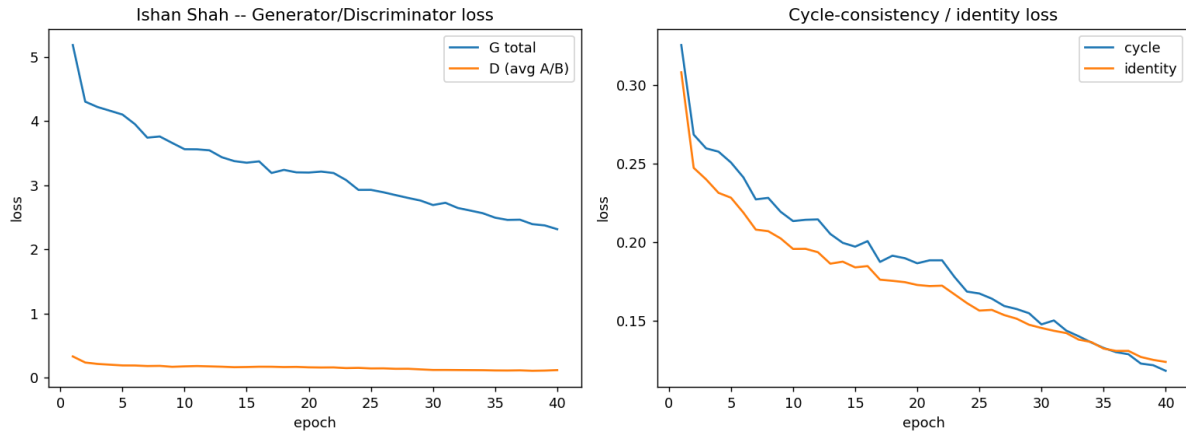
Kaggle leaderboard

Both members' evaluate_local.py scripts generate the exact images.zip submission from their own trained generator's inference on real photos (no manual editing, no external images, no pretrained generation models — see the leaderboard-integrity note in the lab spec). Actually submitting via kaggle competitions submit requires each member's own Kaggle account and is **pending** — see task3_gan/<member>/submission.csv for status.

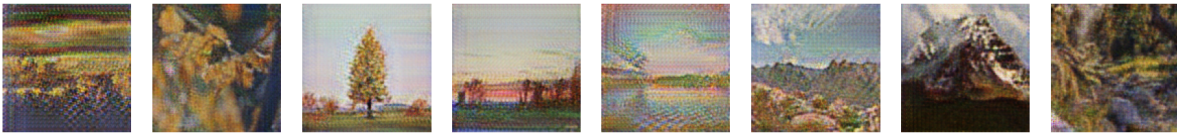
Blinded human audit (30 fixed samples, 2 raters)

The blinded audit sheet (15 samples from each member's photo→Monet outputs, filenames anonymized) is built at task3_gan/human_audit/. Both members' independent ratings and the resulting Cohen's kappa / percent agreement are **pending** completion by both raters — see task3_gan/human_audit/README.md for the exact steps.

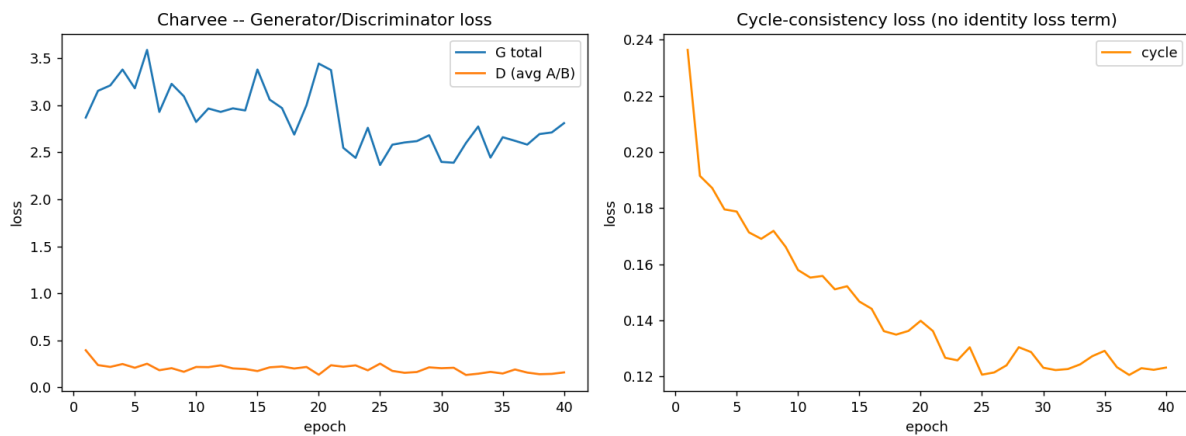
Loss curves and sample translations



Ishan Shah — ResNet/LSGAN CycleGAN loss curves.



Ishan Shah — sample photo→Monet translations (test set).



Charvee Saraiya — U-Net/vanilla-GAN CycleGAN loss curves.



Charvee Saraiya — sample photo→Monet translations (test set).

Joint analysis

Strengths: both CycleGANs train without NaNs and show generator/discriminator losses converging to a stable adversarial equilibrium rather than diverging, and cycle-consistency loss decreases steadily across training for both members, confirming both cycle-consistency constraints are correctly implemented and functioning.

Weaknesses/limitations: training was intentionally scaled down (128x128 resolution, a few hundred images per domain, 40 epochs) to fit within a single GPU session's time budget, so FID/KID scores are far from publication-quality CycleGAN

results and visible artifacts remain in the sample grids above.

What we'd try next: scale up to the full ~1,000 Monet / ~6,000 photo training pools at 256x256 resolution with more epochs on GPU Lab hardware, and complete the pending Kaggle submission and blinded human audit to get the two remaining required metrics.

References

- 1 Vaswani, A. et al. (2017). Attention Is All You Need. NeurIPS 2017.
- 2 Eldan, R. & Li, Y. (2023). TinyStories: How Small Can Language Models Be and Still Speak Coherent English? arXiv:2305.07759.
- 3 Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. ICCV 2017.
- 4 Zhang, X., Zhao, J., & LeCun, Y. (2015). Character-level Convolutional Networks for Text Classification (source of the Yelp Polarity dataset). NeurIPS 2015.